

Problem Set 3

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Invalid Date

Question 1

```
main_data <- read_csv(here("Y2S1/I0/PS3/Data/prod_level_data.csv"))
```

Rows: 71 Columns: 21

-- Column specification -----

Delimiter: ","

dbl (21): market, x1, x2, x3, x4, p1, p2, p3, p4, ave_dist1, ave_dist2, ave...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
main_long <- main_data %>%
  pivot_longer(
    cols = -market,
    names_to = c("var", "product"),
    names_pattern = "([a-z_]+)([1-4])",
    values_to = "value"
  ) %>%
  pivot_wider(
    names_from = var,
    values_from = value
  ) %>%
  mutate(
    product = as.integer(product)
  )

head(main_long)
```

```
# A tibble: 6 x 7
```

```
  market product      x      p ave_dist      s      mc
  <dbl>   <int> <dbl> <dbl>   <dbl> <dbl> <dbl>
1     1     1  2.02  8.71  0.0875 0.48  0.768
2     1     2  0.252  5.20  0.0865 0.098 0.824
3     1     3  3.47   5.10  0.0684 0.163 0.348
4     1     4  0.306  5.03  0.0474 0.12  0.464
5     2     1  0.942  5.43  0.553  0.143 0.721
6     2     2  0.909  4.80  0.966  0.039 0.672
```

```
describe(main_long)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew
market	1	284	36.00	20.53	36.00	36.00	26.69	1.00	71.00	70.00	0.00
product	2	284	2.50	1.12	2.50	2.50	1.48	1.00	4.00	3.00	0.00
x	3	284	2.57	1.48	2.55	2.58	2.00	0.03	4.99	4.96	-0.05
p	4	284	5.46	0.91	5.22	5.33	0.60	4.17	9.56	5.39	1.76
ave_dist	5	284	0.84	0.90	0.55	0.69	0.47	0.01	6.54	6.53	3.21
s	6	284	0.17	0.12	0.15	0.16	0.10	0.01	0.56	0.56	1.10
mc	7	284	0.52	0.29	0.55	0.52	0.35	0.00	1.00	0.99	-0.09

	kurtosis	se
market	-1.21	1.22
product	-1.37	0.07
x	-1.28	0.09
p	3.87	0.05
ave_dist	14.91	0.05
s	0.73	0.01
mc	-1.20	0.02

```
main_graph_pxs <- main_long %>%
  group_by(product) %>%
  ggplot(aes(x = s, y = p)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Price vs. Market Share by Product",
    x = "Market Share (s)",
    y = "Price (p)"
  ) +
  facet_wrap(~ product) +
  theme_minimal()
```

```

main_graph_xxs <- main_long %>%
  group_by(product) %>%
  ggplot(aes(x = s, y = x)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Rating vs. Market Share by Product",
    x = "Rating (x)",
    y = "Market Share (s)"
  ) +
  facet_wrap(~ product) +
  theme_minimal()

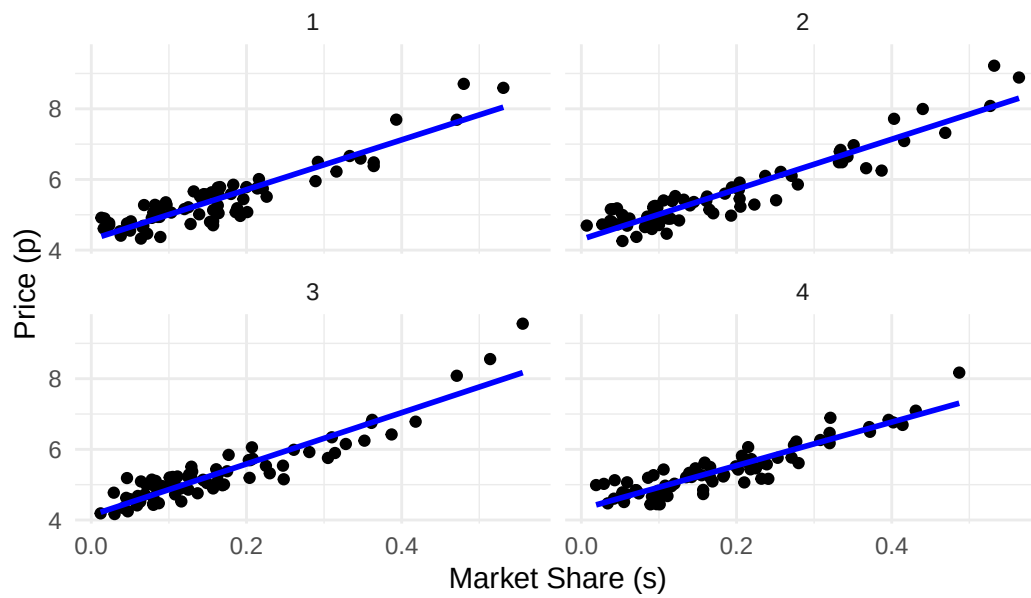
main_graph_sxad <- main_long %>%
  group_by(product) %>%
  ggplot(aes(x = s, y = ave_dist)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Average Distance vs. Market Share by Product",
    x = "Average Distance (ave_dist)",
    y = "Market Share (s)"
  ) +
  facet_wrap(~ product) +
  theme_minimal()

main_graph_pxs

```

`geom\_smooth()` using formula = 'y ~ x'

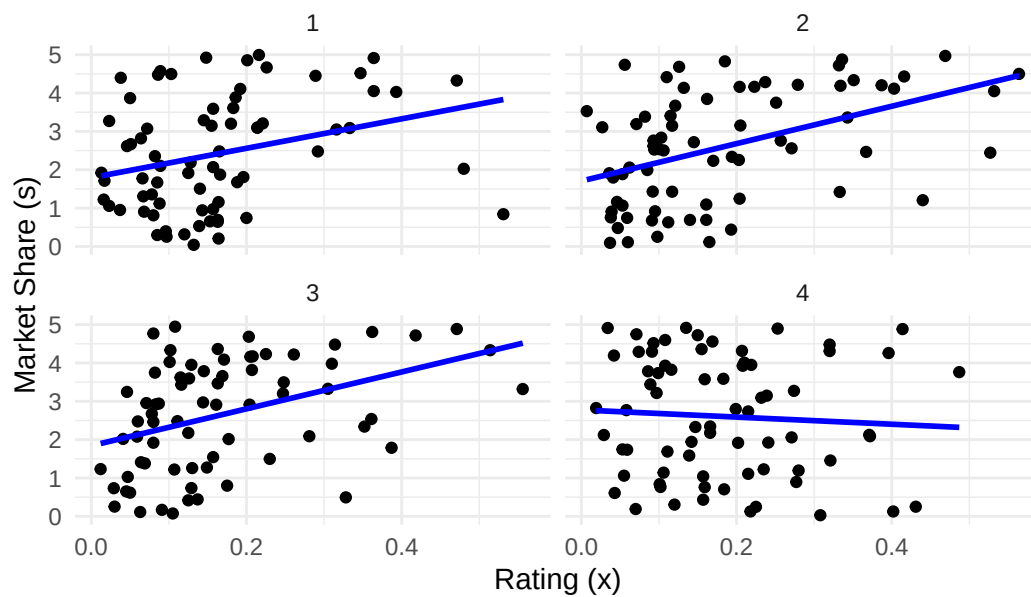
Price vs. Market Share by Product



`main_graph_xxs`

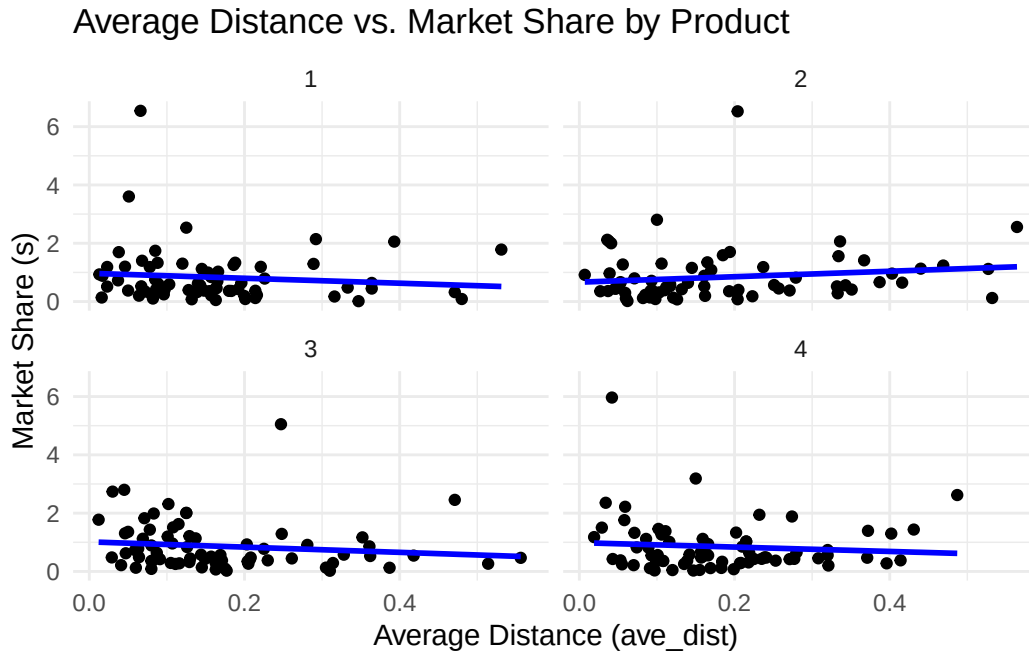
``geom_smooth()`` using formula = `'y ~ x'`

Rating vs. Market Share by Product



```
main_graph_sxad
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Price is positively correlated with market share, with price increasing with share for all products. For Yelp rating, there is no discernable correlation. For distance, there is a slightly negative relationship, though it appears to be as good as flat. All make sense to me, as price is directly manipulable as market power increases, while distance may be harder to control due to real estate effects, and ratings are almost fully out of control.

Question 2

Part 1

```
logit_data <- main_long %>%  
  group_by(market) %>%  
  mutate(  
    s_out = 1 - sum(s),  
    logit_s = log(s) - log(s_out)  
  ) %>%
```

```

ungroup()

mnl1 <- feols(
  logit_s ~ x + p,
  data = logit_data
)
summary(mnl1)

```

OLS estimation, Dep. Var.: logit_s

Observations: 284

Standard-errors: IID

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.744482	0.199805	-23.74556	< 2.2e-16 ***
x	0.059056	0.022765	2.59414	0.0099801 **
p	0.701882	0.037148	18.89406	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.546478 Adj. R2: 0.591078

Part 2

```

logit_data <- logit_data %>%
  group_by(product, market) %>%
  mutate(
    zp = mc
  ) %>%
  ungroup()

mnl2 <- ivreg(
  logit_s ~ x + p | x + zp,
  data = logit_data
)
summary(mnl2)

```

Call:

ivreg(formula = logit_s ~ x + p | x + zp, data = logit_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-3.11303	-0.87252	-0.03998	0.63885	4.13642

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.70956	2.22696	0.768	0.44333
x	0.24623	0.08168	3.015	0.00281 **
p	-0.56814	0.43724	-1.299	0.19488

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.248 on 281 degrees of freedom

Multiple R-Squared: -1.095, Adjusted R-squared: -1.11

Wald test: 6.085 on 2 and 281 DF, p-value: 0.002588

Part 3

```
logit_data <- logit_data %>%
  group_by(product, market) %>%
  mutate(
    ad = ave_dist
  ) %>%
  ungroup()

mnl3 <- ivreg(
  logit_s ~ x + p + ad | x + ad + zp,
  data = logit_data
)
summary(mnl3)
```

Call:

```
ivreg(formula = logit_s ~ x + p + ad | x + ad + zp, data = logit_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.11693	-0.86123	-0.07583	0.65975	4.13945

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept)  1.90340    2.24620    0.847    0.39750
x            0.24942    0.08221    3.034    0.00264 **
p           -0.58279    0.44010   -1.324    0.18651
ad          -0.14442    0.08312   -1.737    0.08341 .

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.255 on 280 degrees of freedom

Multiple R-Squared: -1.111, Adjusted R-squared: -1.134

Wald test: 4.968 on 3 and 280 DF, p-value: 0.002249

In the first model, we see a clear misspecification, since price's coefficient is positive. However, upon instrumenting for price, we flip the sign and get a negative coefficient. But price is not not statistically significant, indicating a weak instrument. Finally, adding distance as an instrument, we see that both price and distance have negative signs, which is to be expected, but neither is statistically significant.

If the true values are the values set for simulation as laid out in part 1, price is biased negatively, distance is slightly biased negatively, and characteristics are slightly biased positively. Thus, all parameters estimated through multinomial logit are overstating the effect of these variables when in the case of complete information.

Question 3

```
search_dat <- read_csv(here("Y2S1/I0/PS3/Data/search_set_data.csv"))
```

Rows: 1136 Columns: 5

-- Column specification -----

Delimiter: ","

dbl (5): market, prod1, prod2, prod3, prod4

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

search_dat_long <- search_dat %>%
  pivot_longer(
    cols = starts_with("prod"),
    names_to = "product",
    names_prefix = "prod",

```



```

    values_to = "in_set"
  ) %>%
  mutate(
    product = as.integer(product)
  ) %>%
  group_by(market) %>%
  mutate(
    set_id = rep(1:16, each = 4)
  ) %>%
  ungroup()

merge_dat <- main_long %>%
  left_join(search_dat_long, by = c("market", "product"))

```

```

# Compute pi_jt to determine entry into search set
pi_fun <- function(mu, ave_dist) {
  p <- plogis(10 + mu * ave_dist)

  p <- pmin(pmax(p, 1e-6), 1 - 1e-6)
  return(p)
}

# Predicting shares

predict_share <- function(theta, main_long, search_dat_long, xi) {
  beta0 <- theta[1]
  beta1 <- theta[2]
  alpha <- theta[3]
  mu <- theta[4]

  main_long <- main_long %>%
    arrange(
      market,
      product
    )

  prod_df <- main_long %>%
    mutate(
      xi = xi,
      delta = beta0 + beta1 * x + alpha * p + xi,
      pi = pi_fun(mu, ave_dist)
    ) %>%

```

```

    select(market, product, delta, pi)

q3_df <- search_dat_long %>%
  arrange(market, set_id, product) %>%
  left_join(prod_df, by = c("market", "product"))
q3_df <- q3_df %>%
  group_by(market, set_id) %>%
  mutate(
    log_Ps = sum(
      in_set * log(pi) + (1 - in_set) * log(1 - pi)
    ),
    P_s = exp(log_Ps),
    denom_S = 1 + sum(if_else(in_set == 1, exp(delta), 0)),
    P_js = if_else(
      in_set == 1,
      exp(delta) / denom_S,
      0
    )
  ) %>%
  ungroup()

pred_shares <- q3_df %>%
  group_by(market, product) %>%
  summarize(
    s_hat = mean(P_s * P_js),
    .groups = "drop"
  ) %>%
  arrange(market, product)

s_hat_vec = pred_shares$s_hat
return(s_hat_vec)
}

# Contraction mapping for xi

xi_cont <- function(theta, main_long, search_dat_long,
                    max_iter = 500, tol = 1e-6) {
  main_long <- main_long %>%
    arrange(market, product)

  # observed shares; clamp zeros if any
  s_obs <- main_long$s

```

```

s_obs <- pmax(s_obs, 1e-12)

# initial xi
xi <- rep(0, nrow(main_long))

for (iter in seq_len(max_iter)) {
  s_hat <- predict_share(theta, main_long, search_dat_long, xi)

  # if predict_shares exploded, bail with NA xi
  if (any(!is.finite(s_hat))) {
    return(rep(NA_real_, length(xi)))
  }

  s_hat <- pmax(s_hat, 1e-12)

  xi_new <- xi + log(s_obs) - log(s_hat)

  # if xi_new exploded, bail
  if (any(!is.finite(xi_new))) {
    return(rep(NA_real_, length(xi)))
  }

  # convergence check
  diff <- max(abs(xi_new - xi))

  if (!is.finite(diff)) {
    # if diff is NA/Inf, bail
    return(rep(NA_real_, length(xi)))
  }

  if (diff < tol) {
    xi <- xi_new
    break
  }

  xi <- xi_new
}

return(xi)
}

# GMM

```

```

GMM_obj <- function(theta, main_long, search_dat_long) {
  main_long <- main_long %>%
    arrange(market, product)

  xi <- xi_cont(theta, main_long, search_dat_long)

  # if contraction failed, slam with big penalty
  if (any(!is.finite(xi))) {
    return(1e12)
  }

  Z <- model.matrix(~ x + mc + ave_dist, data = main_long)

  # if instruments are weird / NA, also slam
  if (any(!is.finite(Z))) {
    return(1e12)
  }

  N <- nrow(Z)
  if (N == 0) {
    return(1e12)
  }

  m <- as.numeric(crossprod(Z, xi)) / N
  if (any(!is.finite(m))) {
    return(1e12)
  }

  Q <- sum(m^2)

  if (!is.finite(Q)) {
    Q <- 1e12
  }

  Q
}

theta_start <- c(
  beta0 = 1,
  beta1 = 0.2,
  alpha = -0.25,
  mu = -0.1
)

```

```
)  
  
gmm_fit <- optim(  
  par = theta_start,  
  fn = GMM_obj,  
  main_long = main_long,  
  search_dat_long = search_dat_long,  
  method = "BFGS",  
  control = list(maxit = 500)  
)  
  
gmm_fit$par
```

```
beta0 beta1 alpha    mu  
  1.00  0.20 -0.25 -0.10
```

```
gmm_fit$value
```

```
[1] 1e+12
```